Table A1

Gender Gap in Numeracy, Literacy, and ICT Literacy Across the School Career (Separate Models)

	Numeracy			Literacy			ICT literacy		
	b	*SE*	*95% CI*	*b*	*SE*	*95% CI*	*b*	*SE*	*95% CI*
Intercept	0.128***	0.018	[0.093, 0.164]	−0.087***	0.021	[−0.127, −0.046]	0.025	0.022	[−0.018, 0.068]
Time (grade)	−0.003	0.003	[−0.008, 0.003]	−0.007*	0.004	[−0.015, 0.000]	0.006	0.004	[−0.002, 0.015]
Gender (ref.: male)									
Female	−0.265***	0.024	[−0.311, −0.219]	0.107***	0.024	[0.060, 0.155]	−0.058*	0.028	[−0.113, −0.003]
Female × Time	0.000	0.003	[−0.007, 0.007]	0.017**	0.005	[0.006, 0.027]	−0.015**	0.005	[−0.026, −0.005]
Cohort (SC3)	−0.030	0.023	[−0.075, 0.014]	−0.010	0.024	[−0.058, 0.038]	−0.029	0.030	[−0.087, 0.029]
SD (random intercept)	0.792	0.007	[0.777, 0.806]	0.745	0.009	[0.727, 0.763]	0.728	0.011	[0.707, 0.749]
SD (residual)	0.582	0.005	[0.572, 0.592]	0.644	0.006	[0.633, 0.657]	0.672	0.009	[0.655, 0.690]
N observations		39,667			26,751			21,385	
N persons		11,971			11,028			10,903	

Note. b = unstandardised coefficient; SE = standard error; CI = confidence interval

*p < .05. **p < .01. ***p < .001.

Table A2

Parental Education Gaps in Numeracy, Literacy, and ICT Literacy Across the School Career (Separate Models)

	Numeracy			Literacy			ICT literacy		
	b	*SE*	*95% CI*	*b*	*SE*	*95% CI*	*b*	*SE*	*95% CI*
Intercept	−0.496***	0.059	[−0.611, −0.380]	−0.570***	0.057	[−0.681, −0.458]	−0.525***	0.073	[−0.669, −0.381]
Time (grade)	0.011	0.008	[−0.004, 0.026]	0.021*	0.009	[0.003, 0.040]	0.012	0.013	[−0.013, 0.037]
Parental education (ref.: up to lower secondary)									
VET qualification	0.339***	0.063	[0.215, 0.463]	0.339***	0.058	[0.224, 0.454]	0.354***	0.076	[0.205, 0.504]
Upper secondary education	0.450***	0.063	[0.326, 0.574]	0.472***	0.062	[0.350, 0.594]	0.488***	0.081	[0.328, 0.647]
Tertiary education	0.755***	0.064	[0.628, 0.881]	0.848***	0.060	[0.731, 0.966]	0.784***	0.076	[0.634, 0.933]
VET qualification × Time	−0.014	0.008	[−0.030, 0.003]	−0.018	0.010	[−0.037, 0.002]	−0.010	0.013	[−0.036, 0.017]
Upper secondary × Time	−0.015	0.008	[−0.031, 0.001]	−0.017	0.011	[−0.039, 0.004]	−0.020	0.014	[−0.048, 0.007]
Tertiary education × Time	−0.015	0.008	[−0.031, 0.000]	−0.031**	0.010	[−0.050, −0.011]	−0.020	0.013	[−0.046, 0.007]
Cohort (SC3)	0.017	0.022	[−0.027, 0.060]	0.054*	0.023	[0.009, 0.099]	0.029	0.029	[−0.028, 0.086]
SD (random intercept)	0.741	0.008	[0.726, 0.756]	0.674	0.010	[0.655, 0.693]	0.678	0.011	[0.656, 0.701]
SD (residual)	0.586	0.005	[0.577, 0.597]	0.652	0.006	[0.640, 0.664]	0.680	0.009	[0.662, 0.698]
N observations		39,667			26,751			21,385	
N persons		11,971			11,028			10,903	

Note. b = unstandardised coefficient; SE = standard error; CI = confidence interval

*p < .05. **p < .01. ***p < .001.

Table A3

Parental Occupation Gaps in Numeracy, Literacy, and ICT Literacy Across the School Career (Separate Models)

	Numeracy			Literacy			ICT literacy		
	b	*SE*	*95% CI*	*b*	*SE*	*95% CI*	*b*	*SE*	*95% CI*
Intercept	−0.351***	0.038	[−0.425, −0.277]	−0.381***	0.037	[−0.454, −0.308]	−0.379***	0.048	[−0.474, −0.285]
Time (grade)	0.007	0.005	[−0.003, 0.018]	0.010	0.007	[−0.004, −0.025]	0.007	0.009	[−0.011, 0.026]
Parental occupation (ref.: manual workers)									
White-collar workers	0.187***	0.047	[0.095, 0.279]	0.149**	0.044	[0.062, 0.236]	0.209**	0.060	[0.091, 0.326]
Lower professionals	0.392***	0.045	[0.304, 0.479]	0.369***	0.042	[0.286, 0.452]	0.393***	0.054	[0.287, 0.498]
Higher professionals and employers	0.513***	0.043	[0.428, 0.597]	0.536***	0.039	[0.459, 0.613]	0.554***	0.052	[0.452, 0.656]
White-collar workers × Time	−0.008	0.006	[−0.020, 0.005]	−0.005	0.008	[−0.021, 0.012]	−0.006	0.011	[−0.027, 0.015]
Lower professionals × Time	−0.014*	0.006	[−0.026, −0.002]	−0.011	0.009	[−0.030, 0.007]	−0.009	0.010	[−0.029, 0.011]
Higher professionals and employers × Time	−0.012*	0.006	[−0.023, 0.000]	−0.016	0.008	[−0.033, 0.000]	−0.015	0.010	[−0.034, 0.003]
Cohort (SC3)	0.007	0.022	[−0.037, 0.050]	0.038	0.024	[−0.008, 0.084]	0.020	0.030	[−0.039, 0.078]
SD (random intercept)	0.758	0.007	[0.744, 0.773]	0.698	0.010	[0.679, 0.718]	0.688	0.011	[0.667, 0.711]
SD (residual)	0.585	0.005	[0.575, 0.595]	0.651	0.006	[0.639, 0.663]	0.679	0.009	[0.662, 0.697]
N observations		39,667			26,751			21,385	
N persons		11,971			11,028			10,903	

Note. b = unstandardised coefficient; SE = standard error; CI = confidence interval

*p < .05. **p < .01. ***p < .001.

Table A4

Migration Background Gaps in Numeracy, Literacy, and ICT Literacy Across the School Career (Separate Models)

	Numeracy			Literacy			ICT literacy		
	b	*SE*	*95% CI*	*b*	*SE*	*95% CI*	*b*	*SE*	*95% CI*
Intercept	0.053***	0.015	[0.023, 0.082]	0.014	0.018	[−0.021, 0.049]	0.050**	0.018	[0.015, 0.085]
Time (grade)	−0.004	0.002	[−0.009, 0.000]	−0.001	0.003	[−0.007, 0.004]	−0.001	0.004	[−0.008, 0.005]
Migration background (ref.: none)									
With migration background	−0.237***	0.031	[−0.297, −0.177]	−0.203***	0.030	[−0.263, −0.144]	−0.233***	0.035	[−0.303, −0.164]
With migration background × Time	0.009*	0.005	[0.000, 0.018]	0.010	0.006	[−0.002, 0.022]	0.001	0.007	[−0.013, 0.016]
Cohort (SC3)	−0.033	0.023	[−0.078, 0.012]	−0.011	0.024	[−0.059, 0.036]	−0.030	0.030	[−0.089, 0.028]
SD (random intercept)	0.791	0.007	[0.776, 0.805]	0.741	0.009	[0.722, 0.759]	0.719	0.011	[0.698, 0.740]
SD (residual)	0.583	0.005	[0.573, 0.593]	0.646	0.006	[0.634, 0.658]	0.674	0.009	[0.657, 0.691]
N observations		39,667			26,751			21,385	
N persons		11,971			11,028			10,903	

Note. b = unstandardised coefficient; SE = standard error; CI = confidence interval

*p < .05. **p < .01. ***p < .001.

Table A5

Mutually Adjusted Models for Numeracy, Literacy, and ICT Literacy

	Numeracy			Literacy			ICT literacy		
	b	*SE*	*95% CI*	*b*	*SE*	*95% CI*	*b*	*SE*	*95% CI*
Intercept	−0.368***	0.069	[−0.503, −0.234]	−0.615***	0.060	[−0.732, −0.498]	−0.529***	0.088	[−0.701, −0.356]
Time (grade)	0.011	0.009	[−0.006, 0.028]	0.005	0.011	[−0.015, 0.026]	0.024	0.014	[−0.004, 0.052]
Gender (ref.: male)									
Female	−0.252***	0.022	[−0.296, −0.209]	0.109***	0.023	[0.064, 0.153]	−0.033	0.027	[−0.086, 0.020]
Female × Time	−0.003	0.003	[−0.010, 0.003]	0.018***	0.005	[0.009, 0.027]	−0.017**	0.005	[−0.027, −0.007]
Parental education (ref.: up to lower secondary)									
VET qualification	0.201**	0.068	[0.068, 0.335]	0.217***	0.060	[0.099, 0.335]	0.218**	0.083	[0.055, 0.381]
Upper secondary education	0.301***	0.069	[0.165, 0.437]	0.328***	0.063	[0.204, 0.453]	0.335***	0.089	[0.160, 0.510]
Tertiary education	0.528***	0.072	[0.387, 0.669]	0.631***	0.069	[0.495, 0.766]	0.566***	0.088	[0.394, 0.738]
VET qualification × Time	−0.008	0.008	[−0.023, 0.008]	−0.009	0.010	[−0.029, 0.010]	−0.007	0.013	[−0.032, 0.019]
Upper secondary × Time	−0.012	0.008	[−0.027, 0.004]	−0.010	0.012	[−0.032, 0.013]	−0.015	0.014	[−0.042, 0.012]
Tertiary education × Time	−0.010	0.008	[−0.026, 0.006]	−0.022	0.012	[−0.045, 0.001]	−0.018	0.014	[−0.045, 0.010]
Parental occupation (ref.: manual workers)									
White-collar workers	0.128*	0.052	[0.026, 0.231]	0.098*	0.047	[0.006, 0.190]	0.146*	0.060	[0.029, 0.263]
Lower professionals	0.242***	0.048	[0.147, 0.337]	0.202***	0.043	[0.117, 0.287]	0.247***	0.059	[0.131, 0.363]
Higher professionals and employers	0.273***	0.051	[0.173, 0.373]	0.248***	0.046	[0.156, 0.339]	0.287***	0.057	[0.175, 0.399]
White-collar workers × Time	−0.005	0.007	[−0.018, 0.008]	−0.003	0.008	[−0.019, 0.014]	−0.005	0.010	[−0.025, 0.014]
Lower professionals × Time	−0.007	0.007	[−0.020, 0.006]	−0.004	0.009	[−0.021, 0.013]	−0.005	0.011	[−0.026, 0.015]
Higher professionals and employers × Time	−0.006	0.007	[−0.020, 0.008]	−0.004	0.009	[−0.022, 0.013]	−0.008	0.010	[−0.028, 0.012]
Migration background (ref.: none)									
With migration background	−0.167***	0.031	[−0.228, −0.106]	−0.124***	0.029	[−0.181, −0.067]	−0.141***	0.038	[−0.215, −0.067]
With migration background × Time	0.006	0.005	[−0.003, 0.015]	0.003	0.006	[−0.008, 0.015]	−0.003	0.007	[−0.016, 0.010]
Cohort (SC3)	0.016	0.021	[−0.025, 0.058]	0.061**	0.023	[0.016, 0.106]	0.029	0.029	[−0.028, 0.087]
SD (random intercept)	0.716	0.007	[0.702, 0.730]	0.659	0.010	[0.640, 0.678]	0.661	0.011	[0.640, 0.683]
SD (residual)	0.581	0.005	[0.571, 0.591]	0.650	0.006	[0.637, 0.662]	0.679	0.008	[0.662, 0.696]
N observations		45,652			32,736			25,451	
N persons		13,890			12,947			12,822	

Note. b = unstandardised coefficient; SE = standard error; CI = confidence interval

*p < .05. **p < .01. ***p < .001.

Table A6

Gaps by Region of Origin (Five Categories) in Numeracy, Literacy, and ICT Literacy Without Covariates

	Numeracy			Literacy			ICT literacy		
	b	*SE*	*95% CI*	*b*	*SE*	*95% CI*	*b*	*SE*	*95% CI*
Intercept	0.066***	0.016	[0.035, 0.097]	0.020	0.018	[−0.015, 0.055]	0.051**	0.018	[0.016, 0.087]
Time (grade)	−0.011**	0.004	[−0.018, −0.004]	−0.006	0.005	[−0.016, 0.004]	−0.001	0.006	[−0.013, 0.011]
Region of origin (ref.: German)									
Western Europe	−0.133	0.075	[−0.281, 0.015]	−0.034	0.075	[−0.181, 0.113]	−0.126	0.090	[−0.302, 0.050]
Eastern Europe	−0.223***	0.063	[−0.347, −0.100]	−0.168*	0.066	[−0.298, −0.037]	−0.161*	0.074	[−0.306, −0.016]
FSU	−0.248***	0.056	[−0.358, −0.139]	−0.233***	0.056	[−0.343, −0.123]	−0.237***	0.066	[−0.367, −0.107]
Turkey/MENA	−0.424***	0.058	[−0.538, −0.310]	−0.442***	0.057	[−0.553, −0.331]	−0.429***	0.079	[−0.584, −0.275]
Other regions	−0.286***	0.058	[−0.400, −0.171]	−0.141	0.087	[−0.312, 0.030]	−0.137	0.134	[−0.400, 0.126]
Western Europe × Time	0.009	0.015	[−0.021, 0.039]	−0.017	0.022	[−0.060, 0.027]	−0.005	0.023	[−0.049, 0.040]
Eastern Europe × Time	0.010	0.012	[−0.013, 0.034]	0.012	0.020	[−0.028, 0.052]	−0.005	0.018	[−0.040, 0.030]
FSU × Time	0.010	0.011	[−0.011, 0.031]	0.010	0.018	[−0.026, 0.046]	−0.015	0.018	[−0.051, 0.021]
Turkey/MENA × Time	0.019	0.010	[−0.001, 0.040]	0.037*	0.017	[0.004, 0.070]	0.002	0.019	[−0.035, 0.039]
Other regions × Time	0.027*	0.011	[0.005, 0.049]	0.011	0.022	[−0.033, 0.054]	−0.003	0.029	[−0.060, 0.054]
Cohort (SC3)	−0.026	0.025	[−0.075, 0.024]	−0.009	0.026	[−0.061, 0.042]	−0.040	0.035	[−0.110, 0.029]
SD (random intercept)	0.785	0.008	[0.771, 0.800]	0.753	0.009	[0.735, 0.772]	0.749	0.012	[0.726, 0.772]
SD (residual)	0.582	0.005	[0.572, 0.592]	0.623	0.008	[0.607, 0.640]	0.630	0.011	[0.608, 0.652]
N observations		38,480			25,568			18,285	
N persons		13,887			12,945			12,819	

Note. b = unstandardised coefficient; SE = standard error; CI = confidence interval

*p < .05. **p < .01. ***p < .001.

Table A7

Gaps by Region of Origin (Five Categories) in Numeracy, Literacy, and ICT Literacy Adjusted for Covariates

	Numeracy			Literacy			ICT literacy		
	b	*SE*	*95% CI*	*b*	*SE*	*95% CI*	*b*	*SE*	*95% CI*
Intercept	−0.304***	0.070	[−0.442, −0.166]	−0.699***	0.071	[−0.838, −0.560]	−0.443***	0.092	[−0.624, −0.262]
Time (grade)	−0.022	0.013	[−0.048, 0.003]	0.026	0.020	[−0.014, 0.065]	−0.016	0.022	[−0.059, 0.027]
Gender (ref.: male)									
Female	−0.314***	0.025	[−0.364, −0.265]	0.129***	0.024	[0.081, 0.177]	−0.080**	0.028	[−0.136, −0.024]
Female × Time	0.016**	0.005	[0.007, 0.026]	0.007	0.008	[−0.009, 0.022]	0.003	0.007	[−0.011, 0.017]
Parental education (ref.: up to lower secondary)									
VET qualification	0.190**	0.066	[0.061, 0.319]	0.271***	0.071	[0.132, 0.410]	0.164	0.089	[−0.011, 0.339]
Upper secondary education	0.277***	0.070	[0.141, 0.413]	0.383***	0.073	[0.241, 0.526]	0.260**	0.095	[0.074, 0.445]
Tertiary education	0.500***	0.071	[0.361, 0.640]	0.693***	0.079	[0.537, 0.849]	0.461***	0.094	[0.277, 0.644]
VET qualification × Time	0.000	0.012	[−0.023, 0.023]	−0.029	0.020	[−0.067, 0.010]	0.014	0.019	[−0.023, 0.051]
Upper secondary × Time	0.003	0.012	[−0.021, 0.027]	−0.022	0.021	[−0.062, 0.019]	0.018	0.021	[−0.022, 0.058]
Tertiary education × Time	0.011	0.012	[−0.012, 0.035]	−0.036	0.023	[−0.081, 0.009]	0.031	0.021	[−0.010, 0.072]
Parental occupation (ref.: manual workers)									
White-collar workers	0.132*	0.052	[0.028, 0.235]	0.097	0.054	[−0.010, 0.204]	0.160*	0.064	[0.035, 0.285]
Lower professionals	0.242***	0.052	[0.141, 0.344]	0.227***	0.054	[0.120, 0.333]	0.257***	0.065	[0.130, 0.385]
Higher professionals and employers	0.264***	0.054	[0.157, 0.370]	0.280***	0.058	[0.166, 0.394]	0.309***	0.066	[0.179, 0.439]
White-collar workers × Time	−0.003	0.008	[−0.020, 0.014]	0.001	0.015	[−0.027, 0.030]	−0.006	0.014	[−0.034, 0.022]
Lower professionals × Time	−0.002	0.009	[−0.019, 0.016]	−0.006	0.016	[−0.037, 0.025]	−0.001	0.015	[−0.030, 0.028]
Higher professionals and employers × Time	0.003	0.009	[−0.015, 0.021]	−0.012	0.017	[−0.046, 0.022]	−0.008	0.016	[−0.040, 0.024]
Region of origin (ref.: German)									
Western Europe	−0.110	0.073	[−0.254, 0.034]	0.002	0.073	[−0.141, 0.146]	−0.107	0.087	[−0.278, 0.065]
Eastern Europe	−0.162**	0.060	[−0.280, −0.043]	−0.068	0.062	[−0.189, 0.053]	−0.078	0.073	[−0.222, 0.065]
FSU	−0.171**	0.054	[−0.276, −0.065]	−0.103	0.056	[−0.213, 0.006]	−0.132*	0.064	[−0.258, −0.006]
Turkey/MENA	−0.349***	0.057	[−0.462, −0.237]	−0.310***	0.054	[−0.416, −0.203]	−0.293***	0.082	[−0.454, −0.132]
Other regions	−0.229***	0.057	[−0.340, −0.118]	−0.092	0.084	[−0.257, 0.073]	−0.087	0.132	[−0.347, 0.173]
Western Europe × Time	0.011	0.015	[−0.018, 0.041]	−0.018	0.023	[−0.062, 0.027]	−0.001	0.022	[−0.044, 0.042]
Eastern Europe × Time	0.011	0.012	[−0.012, 0.035]	0.004	0.020	[−0.036, 0.043]	−0.002	0.018	[−0.037, 0.033]
FSU × Time	0.012	0.011	[−0.010, 0.033]	−0.003	0.018	[−0.039, 0.033]	−0.009	0.018	[−0.044, 0.026]
Turkey/MENA × Time	0.020	0.010	[0.000, 0.041]	0.022	0.017	[−0.012, 0.055]	0.001	0.020	[−0.037, 0.040]
Other regions × Time	0.025*	0.011	[0.002, 0.047]	0.008	0.022	[−0.035, 0.050]	−0.003	0.028	[−0.058, 0.052]
Cohort (SC3)	0.043	0.024	[−0.004, 0.090]	0.077**	0.025	[0.028, 0.126]	0.027	0.036	[−0.044, 0.098]
SD (random intercept)	0.704	0.008	[0.690, 0.720]	0.673	0.010	[0.653, 0.693]	0.688	0.012	[0.664, 0.713]
SD (residual)	0.587	0.005	[0.577, 0.598]	0.633	0.008	[0.616, 0.649]	0.641	0.012	[0.618, 0.664]
N observations		38,480			25,568			18,285	
N persons		13,887			12,945			12,819	

Note. b = unstandardised coefficient; SE = standard error; CI = confidence interval

*p < .05. **p < .01. ***p < .001.

Table A8

Gaps by Region of Origin With Turkey/MENA as Reference Group in Numeracy, Literacy, and ICT Literacy Without Covariates (Supplementary Models)

	Numeracy			Literacy			ICT literacy		
	b	*SE*	*95% CI*	*b*	*SE*	*95% CI*	*b*	*SE*	*95% CI*
Intercept	−0.477***	0.055	[−0.585, −0.370]	−0.441***	0.059	[−0.556, −0.325]	−0.425***	0.080	[−0.583, −0.267]
Time (grade)	0.011	0.011	[−0.010, 0.033]	0.027	0.017	[−0.006, 0.060]	−0.012	0.020	[−0.051, 0.028]
Region of origin (ref.: Turkey/MENA)									
Western Europe	0.332***	0.094	[0.147, 0.516]	0.428***	0.088	[0.256, 0.600]	0.311**	0.116	[0.083, 0.540]
Eastern Europe	0.231**	0.082	[0.071, 0.391]	0.286**	0.085	[0.120, 0.452]	0.281*	0.109	[0.066, 0.496]
FSU	0.201**	0.075	[0.053, 0.349]	0.220**	0.077	[0.069, 0.371]	0.210*	0.102	[0.009, 0.411]
Other regions	0.171*	0.074	[0.026, 0.316]	0.308**	0.104	[0.103, 0.512]	0.264	0.163	[−0.056, 0.583]
Western Europe × Time	−0.011	0.019	[−0.048, 0.027]	−0.048	0.026	[−0.100, 0.003]	−0.007	0.027	[−0.060, 0.046]
Eastern Europe × Time	−0.008	0.016	[−0.039, 0.023]	−0.023	0.027	[−0.075, 0.029]	−0.007	0.026	[−0.058, 0.043]
FSU × Time	−0.011	0.015	[−0.040, 0.018]	−0.026	0.024	[−0.073, 0.022]	−0.019	0.027	[−0.071, 0.033]
Other regions × Time	0.006	0.014	[−0.022, 0.034]	−0.022	0.028	[−0.077, 0.033]	0.005	0.035	[−0.063, 0.073]
Cohort (SC3)	−0.080	0.054	[−0.186, 0.025]	−0.096	0.058	[−0.209, 0.017]	0.021	0.085	[−0.145, 0.187]
SD (random intercept)	0.762	0.016	[0.731, 0.795]	0.742	0.021	[0.702, 0.784]	0.782	0.026	[0.733, 0.834]
SD (residual)	0.576	0.010	[0.556, 0.596]	0.617	0.014	[0.589, 0.646]	0.613	0.023	[0.569, 0.661]
N observations		8,403			5,359			3,869	
N persons		3,705			3,212			2,973	

Note. b = unstandardised coefficient; SE = standard error; CI = confidence interval

*p < .05. **p < .01. ***p < .001.

Table A9

Gaps by Region of Origin With Turkey/MENA as Reference Group in Numeracy, Literacy, and ICT Literacy Adjusted for Covariates (Supplementary Models)

	Numeracy			Literacy			ICT literacy		
	b	*SE*	*95% CI*	*b*	*SE*	*95% CI*	*b*	*SE*	*95% CI*
Intercept	−0.781***	0.088	[−0.954, −0.607]	−1.009***	0.095	[−1.195, −0.824]	−0.801***	0.113	[−1.023, −0.578]
Time (grade)	0.004	0.017	[−0.029, 0.037]	0.036	0.026	[−0.015, 0.087]	−0.023	0.028	[−0.077, 0.032]
Gender (ref.: male)									
Female	−0.312***	0.055	[−0.419, −0.205]	0.138*	0.055	[0.031, 0.245]	−0.030	0.067	[−0.162, 0.101]
Female × Time	0.018	0.010	[−0.001, 0.037]	0.008	0.017	[−0.026, 0.041]	0.010	0.016	[−0.021, 0.042]
Parental education (ref.: up to lower secondary)									
VET qualification	0.191*	0.085	[0.025, 0.358]	0.228**	0.081	[0.069, 0.387]	0.137	0.115	[−0.089, 0.362]
Upper secondary education	0.296**	0.091	[0.117, 0.476]	0.313**	0.097	[0.122, 0.503]	0.194	0.116	[−0.034, 0.422]
Tertiary education	0.526***	0.105	[0.320, 0.732]	0.713***	0.118	[0.482, 0.944]	0.435**	0.131	[0.178, 0.691]
VET qualification × Time	−0.004	0.016	[−0.035, 0.028]	−0.019	0.026	[−0.070, 0.032]	0.006	0.027	[−0.047, 0.058]
Upper secondary × Time	0.006	0.017	[−0.028, 0.040]	−0.004	0.028	[−0.058, 0.050]	0.026	0.026	[−0.026, 0.077]
Tertiary education × Time	0.023	0.021	[−0.018, 0.063]	−0.023	0.037	[−0.097, 0.050]	0.042	0.034	[−0.024, 0.108]
Parental occupation (ref.: manual workers)									
White-collar workers	0.171*	0.082	[0.011, 0.331]	0.081	0.090	[−0.095, 0.257]	0.201	0.152	[−0.098, 0.501]
Lower professionals	0.266**	0.085	[0.100, 0.433]	0.240**	0.089	[0.066, 0.415]	0.321*	0.153	[0.021, 0.622]
Higher professionals and employers	0.300**	0.089	[0.124, 0.475]	0.365***	0.103	[0.162, 0.568]	0.444**	0.168	[0.115, 0.773]
White-collar workers × Time	−0.013	0.014	[−0.041, 0.015]	0.008	0.026	[−0.043, 0.058]	−0.013	0.025	[−0.063, 0.037]
Lower professionals × Time	−0.010	0.016	[−0.040, 0.021]	−0.019	0.025	[−0.068, 0.031]	−0.020	0.027	[−0.073, 0.033]
Higher professionals and employers × Time	0.003	0.018	[−0.032, 0.039]	−0.014	0.028	[−0.069, 0.041]	−0.023	0.029	[−0.079, 0.033]
Region of origin (ref.: Turkey/MENA)									
Western Europe	0.267**	0.093	[0.085, 0.449]	0.301**	0.087	[0.131, 0.472]	0.177	0.119	[−0.057, 0.411]
Eastern Europe	0.210**	0.078	[0.057, 0.364]	0.246**	0.078	[0.094, 0.399]	0.224*	0.108	[0.013, 0.436]
FSU	0.202**	0.073	[0.059, 0.344]	0.209**	0.072	[0.067, 0.350]	0.172	0.100	[−0.024, 0.368]
Other regions	0.152*	0.074	[0.006, 0.297]	0.213*	0.104	[0.009, 0.416]	0.164	0.166	[−0.162, 0.489]
Western Europe × Time	−0.012	0.019	[−0.049, 0.025]	−0.035	0.026	[−0.087, 0.017]	−0.002	0.027	[−0.056, 0.051]
Eastern Europe × Time	−0.009	0.016	[−0.040, 0.023]	−0.017	0.026	[−0.067, 0.034]	−0.003	0.026	[−0.054, 0.047]
FSU × Time	−0.010	0.015	[−0.040, 0.019]	−0.023	0.024	[−0.070, 0.024]	−0.011	0.026	[−0.063, 0.040]
Other regions × Time	0.000	0.015	[−0.029, 0.030]	−0.011	0.029	[−0.068, 0.045]	0.005	0.035	[−0.064, 0.074]
Cohort (SC3)	0.007	0.050	[−0.092, 0.105]	0.015	0.057	[−0.098, 0.127]	0.112	0.089	[−0.062, 0.285]
SD (random intercept)	0.671	0.017	[0.639, 0.705]	0.651	0.023	[0.607, 0.699]	0.719	0.027	[0.668, 0.773]
SD (residual)	0.581	0.010	[0.561, 0.601]	0.624	0.015	[0.596, 0.654]	0.623	0.025	[0.576, 0.674]
N observations		8,403			5,359			3,869	
N persons		3,705			3,212			2,973	

Note. b = unstandardised coefficient; SE = standard error; CI = confidence interval

*p < .05. **p < .01. ***p < .001.

Table A10

Gaps by Region of Origin (Six Categories) in Numeracy, Literacy, and ICT Literacy Without Covariates

	Numeracy			Literacy			ICT literacy		
	b	*SE*	*95% CI*	*b*	*SE*	*95% CI*	*b*	*SE*	*95% CI*
Intercept	0.066***	0.016	[0.035, 0.097]	0.020	0.018	[−0.015, 0.055]	0.051**	0.018	[0.016, 0.087]
Time (grade)	−0.011**	0.004	[−0.018, −0.004]	−0.006	0.005	[−0.016, 0.003]	−0.001	0.006	[−0.013, 0.011]
Region of origin (ref.: German)									
Western Europe	−0.133	0.075	[−0.281, 0.015]	−0.034	0.075	[−0.182, 0.113]	−0.126	0.090	[−0.302, 0.050]
Eastern Europe	−0.223***	0.063	[−0.347, −0.100]	−0.168*	0.066	[−0.298, −0.038]	−0.161*	0.074	[−0.306, −0.016]
FSU	−0.248***	0.056	[−0.357, −0.138]	−0.234***	0.056	[−0.344, −0.123]	−0.237***	0.066	[−0.367, −0.107]
Turkey	−0.432***	0.069	[−0.568, −0.296]	−0.465***	0.062	[−0.587, −0.344]	−0.447***	0.090	[−0.625, −0.270]
Other regions	−0.285***	0.058	[−0.400, −0.171]	−0.142	0.087	[−0.313, 0.029]	−0.138	0.134	[−0.401, 0.125]
MENA	−0.407***	0.110	[−0.623, −0.192]	−0.355**	0.131	[−0.612, −0.097]	−0.37*	0.157	[−0.682, −0.067]
Western Europe × Time	0.009	0.015	[−0.021, 0.039]	−0.017	0.022	[−0.060, 0.027]	−0.005	0.023	[−0.049, 0.040]
Eastern Europe × Time	0.010	0.012	[−0.013, 0.034]	0.012	0.020	[−0.028, 0.052]	−0.005	0.018	[−0.040, 0.030]
FSU × Time	0.010	0.011	[−0.011, 0.031]	0.010	0.018	[−0.026, 0.046]	−0.015	0.018	[−0.051, 0.021]
Turkey × Time	0.017	0.012	[−0.007, 0.041]	0.038*	0.018	[0.003, 0.073]	0.000	0.021	[−0.041, 0.040]
Other regions × Time	0.027*	0.011	[0.005, 0.048]	0.011	0.022	[−0.033, 0.054]	−0.003	0.029	[−0.060, 0.054]
MENA × Time	0.032	0.024	[−0.015, 0.078]	0.040	0.044	[−0.046, 0.126]	0.021	0.042	[−0.062, 0.104]
Cohort (SC3)	−0.025	0.025	[−0.074, 0.024]	−0.009	0.026	[−0.060, 0.043]	−0.040	0.035	[−0.109, 0.030]
SD (random intercept)	0.785	0.008	[0.770, 0.800]	0.753	0.009	[0.735, 0.772]	0.748	0.012	[0.725, 0.772]
SD (residual)	0.582	0.005	[0.572, 0.592]	0.623	0.008	[0.607, 0.640]	0.630	0.011	[0.608, 0.652]
N observations		38,480			25,568			18,285	
N persons		13,887			12,945			12,819	

Note. b = unstandardised coefficient; SE = standard error; CI = confidence interval

*p < .05. **p < .01. ***p < .001.

Table A11

Gaps by Region of Origin (Six Categories) in Numeracy, Literacy, and ICT Literacy Adjusted for Covariates

	Numeracy			Literacy			ICT literacy		
	b	*SE*	*95% CI*	*b*	*SE*	*95% CI*	*b*	*SE*	*95% CI*
Intercept	−0.304***	0.071	[−0.442, −0.166]	−0.699***	0.071	[−0.838, −0.560]	−0.443***	0.092	[−0.625, −0.262]
Time (grade)	−0.022	0.013	[−0.048, 0.004]	0.026	0.020	[−0.014, 0.065]	−0.015	0.022	[−0.058, 0.028]
Gender (ref.: male)									
Female	−0.314***	0.025	[−0.364, −0.265]	0.129***	0.024	[0.082, 0.177]	−0.080**	0.028	[−0.136, −0.024]
Female × Time	0.016**	0.005	[0.007, 0.026]	0.007	0.008	[−0.009, 0.022]	0.003	0.007	[−0.011, 0.017]
Parental education (ref.: up to lower secondary)									
VET qualification	0.190**	0.066	[0.061, 0.319]	0.271***	0.070	[0.133, 0.409]	0.164	0.089	[−0.012, 0.339]
Upper secondary education	0.277***	0.070	[0.140, 0.414]	0.383***	0.073	[0.241, 0.526]	0.260**	0.095	[0.074, 0.446]
Tertiary education	0.500***	0.071	[0.361, 0.640]	0.693***	0.079	[0.537, 0.848]	0.461***	0.093	[0.278, 0.644]
VET qualification × Time	0.000	0.012	[−0.023, 0.023]	−0.029	0.020	[−0.067, 0.010]	0.014	0.019	[−0.023, 0.051]
Upper secondary × Time	0.003	0.012	[−0.021, 0.027]	−0.021	0.021	[−0.062, 0.019]	0.018	0.021	[−0.022, 0.058]
Tertiary education × Time	0.011	0.012	[−0.012, 0.035]	−0.036	0.023	[−0.081, 0.009]	0.031	0.021	[−0.010, 0.072]
Parental occupation (ref.: manual workers)									
White-collar workers	0.132*	0.053	[0.029, 0.235]	0.093	0.091	[−0.086, 0.272]	0.173*	0.087	[0.003, 0.343]
Lower professionals	0.243***	0.052	[0.141, 0.344]	0.243*	0.095	[0.056, 0.431]	0.260**	0.090	[0.084, 0.436]
Higher professionals and employers	0.264***	0.054	[0.157, 0.371]	0.314**	0.103	[0.111, 0.518]	0.326**	0.093	[0.143, 0.509]
White-collar workers × Time	−0.003	0.008	[−0.020, 0.013]	0.001	0.015	[−0.027, 0.030]	−0.006	0.014	[−0.035, 0.022]
Lower professionals × Time	−0.002	0.009	[−0.019, 0.015]	−0.006	0.016	[−0.037, 0.025]	−0.001	0.015	[−0.030, 0.028]
Higher professionals and employers × Time	0.003	0.009	[−0.015, 0.021]	−0.012	0.017	[−0.046, 0.022]	−0.008	0.016	[−0.040, 0.023]
Region of origin (ref.: German)									
Western Europe	−0.110	0.073	[−0.254, 0.033]	0.002	0.073	[−0.142, 0.146]	−0.107	0.087	[−0.278, 0.065]
Eastern Europe	−0.162**	0.060	[−0.280, −0.043]	−0.068	0.062	[−0.189, 0.053]	−0.078	0.073	[−0.221, 0.065]
FSU	−0.170**	0.054	[−0.276, −0.065]	−0.104	0.056	[−0.213, 0.006]	−0.132*	0.064	[−0.258, −0.006]
Turkey	−0.348***	0.067	[−0.480, −0.216]	−0.321***	0.061	[−0.440, −0.201]	−0.289**	0.096	[−0.477, −0.101]
Other regions	−0.229***	0.057	[−0.340, −0.118]	−0.092	0.084	[−0.257, 0.073]	−0.087	0.132	[−0.347, 0.173]
MENA	−0.356**	0.107	[−0.565, −0.147]	−0.271*	0.121	[−0.509, −0.034]	−0.304*	0.148	[−0.594, −0.013]
Western Europe × Time	0.011	0.015	[−0.018, 0.041]	−0.018	0.023	[−0.062, 0.027]	−0.001	0.022	[−0.044, 0.042]
Eastern Europe × Time	0.011	0.012	[−0.012, 0.035]	0.004	0.020	[−0.036, 0.043]	−0.002	0.018	[−0.037, 0.032]
FSU × Time	0.012	0.011	[−0.010, 0.033]	−0.003	0.018	[−0.039, 0.033]	−0.009	0.018	[−0.044, 0.026]
Turkey × Time	0.018	0.012	[−0.005, 0.042]	0.021	0.019	[−0.017, 0.058]	−0.003	0.022	[−0.046, 0.041]
Other regions × Time	0.025*	0.011	[0.002, 0.047]	0.008	0.022	[−0.035, 0.050]	−0.003	0.028	[−0.058, 0.052]
MENA × Time	0.030	0.023	[−0.015, 0.076]	0.031	0.043	[−0.053, 0.115]	0.021	0.042	[−0.062, 0.103]
Cohort (SC3)	0.043	0.024	[−0.004, 0.090]	0.077**	0.025	[0.028, 0.127]	0.027	0.036	[−0.044, 0.097]
SD (random intercept)	0.704	0.008	[0.690, 0.720]	0.673	0.010	[0.653, 0.693]	0.688	0.012	[0.664, 0.713]
SD (residual)	0.587	0.005	[0.577, 0.598]	0.632	0.008	[0.616, 0.649]	0.641	0.012	[0.618, 0.664]
N observations		38,480			25,568			18,285	
N persons		13,887			12,945			12,819	

Note. b = unstandardised coefficient; SE = standard error; CI = confidence interval

*p < .05. **p < .01. ***p < .001.